

Yes. I'm taking the flowchart you uploaded as the **base master flow** and expanding it without removing any of the existing ideas. I'm only adding the things we discussed later: **RAG regulatory engine, master project dossier/common application layer, documentation guidance, environmental classification, verified-data provenance, payments, cross-form consistency, risk-based scrutiny, desk-by-desk tracking, consolidated queries, delta re-scrutiny, regulatory-change impact, business-change simulation, approval-to-compliance generation, officer-side RAG, deeper bottleneck analytics, expansion/amendment lifecycle, and incentive lifecycle improvements.**

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MAHARASHTRA INDUSTRIAL APPROVAL, COMPLIANCE & REGULATORY INTELLIGENCE SYSTEM

COMPLETE A-Z MASTER FEATURE FLOW

MAHARASHTRA INDUSTRIAL APPROVAL, COMPLIANCE & INTELLIGENCE SYSTEM
COMPLETE END-TO-END MASTER FLOW

ENTREPRENEUR



1. MAITRI REGISTRATION
- Create entrepreneur / organisation account
 - Contact details
 - Authentication
 - Identity / business-user verification
 - Create one or multiple business projects



2. CREATE BUSINESS / PROJECT

Example:

"Cattle Feed Manufacturing Unit - Ratnagiri"

OR

"Pharmaceutical Manufacturing Unit - Thane"

OR

"Auto Component Factory - Pune"



3. BASIC BUSINESS REQUIREMENTS

Entrepreneur selects only what they already know:

- Need land?
- Existing land?
- MIDC / Non-MIDC?
- New construction?
- Need water?
- Need power?
- Manufacturing / trading / services?
- New unit / existing unit / expansion?
- Existing approvals, if any?

IMPORTANT:

Entrepreneur is NOT expected to know all departments, approvals or NOCs.



BUSINESS DISCOVERY ENGINE



4. ADAPTIVE BUSINESS QUESTIONNAIRE

System asks only relevant questions based on earlier answers.

CORE PARAMETERS

- Industry / sector
- Business activity
- Products / services
- Maharashtra district / location
- Taluka / local jurisdiction
- MIDC / Non-MIDC
- Plot / land status
- Land type
- Investment
- Fixed capital investment
- Number of workers
- Production capacity
- Water requirement
- Wastewater generation

- Air emissions
- Hazardous substances
- Hazardous waste
- Boiler
- Pressure vessel
- New building
- Building area
- Building height / occupancy where relevant
- Power requirement
- Storage activities
- Import / export
- Machinery / dangerous processes
- Project stage
- Existing approvals
- Relevant entrepreneur details for incentives

CONDITIONAL QUESTION TREE

Example:

Hazardous Chemical = YES

↓

Ask chemical quantity / type / storage questions

↓

Additional safety / environment requirements may appear

Hazardous Chemical = NO

↓

Those questions and requirements disappear



5. MASTER BUSINESS PROFILE

Creates the reusable identity of the business.

- Company name
- PAN / CIN
- Promoters / authorised persons
- Address
- District
- Plot
- Industry
- Activity
- Products
- Investment
- Employees
- Production
- Water requirement
- Power requirement
- Pollution attributes

- Safety attributes
- Project stage

This data is reused across every approval, compliance and incentive flow.



MASTER PROJECT DOSSIER / COMMON DATA LAYER



6. COMMON PROJECT DOSSIER

The entrepreneur does NOT fill the same project information repeatedly.

Stores:

- Business details
- Project details
- Promoter details
- Land information
- Building information
- Investment
- Employment
- Production capacity
- Process
- Machinery
- Water
- Power
- Environmental information
- Safety / hazardous-process information

Every departmental form pulls reusable fields from this dossier.

Department-specific forms ask ONLY the additional information required.



VERIFIED DATA PROVENANCE LAYER



7. EVERY IMPORTANT DATA FIELD STORES

VALUE

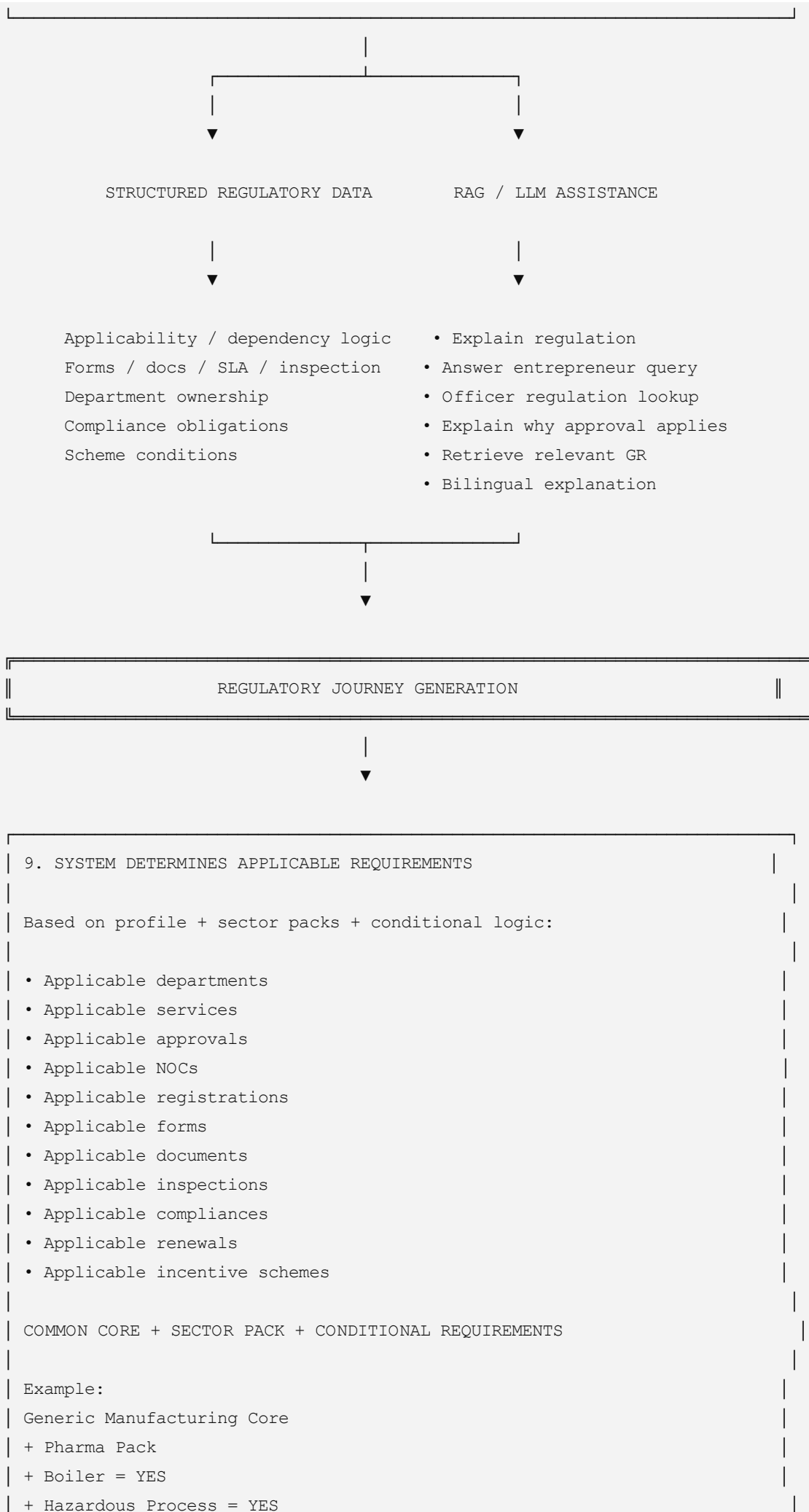
| + SOURCE
| + VERIFICATION STATUS
| + ISSUE DATE
| + EXPIRY DATE
| + WHERE IT IS USED
|
| Example:
|
| Plot Area = 4,800 m²
| Source = MIDC allotment document
| Status = Verified
| Used by:
| • MPCB CTE
| • Building Plan
| • Fire
| • DISH
|
| Production Capacity = 100 T/day
| Source = Entrepreneur declaration
| Status = Self-declared
|



|| REGULATORY KNOWLEDGE ENGINE ||



| 8. EXISTING MAHARASHTRA RAG PIPELINE IS INTEGRATED AS AN API
|
| Inputs / corpus:
| • Government Resolutions
| • Acts
| • Rules
| • Circulars
| • Departmental guidelines
| • Approval descriptions
| • Forms
| • Policy documents
|
| Existing capabilities reused:
| • Neo4j knowledge graph
| • Semantic retrieval
| • Cosine similarity
| • Relevant phrase / clause retrieval
| • Bilingual English + Marathi support
|
| RAG is NOT the legal approval authority.
| It understands, retrieves, explains and assists.
|



| = Custom regulatory journey |



| 10. ENVIRONMENTAL / REGULATORY CLASSIFICATION |

| Where applicable, business characteristics also determine: |

- | • Pollution-related category / classification |
- | • Need for environmental approvals |
- | • Need for enhanced scrutiny |
- | • Need for additional environmental documents |
- | • Need for inspection |

| Classification becomes an INPUT to the approval journey and risk routing. |



| 11. CLASSIFY REQUIREMENTS |

- | A. Confirmed Applicable |
- | B. Conditional |
- | C. Needs Verification |
- | D. Not Applicable |

| "Needs Verification" handles unusual or legally ambiguous cases. |

| It prevents the system from pretending AI knows every legal answer. |



| 12. IDENTIFY REQUIRED SERVICES / FORMS / DOCUMENTS |

| Department |



| Service |



| Required Form(s) |



| Required Documents |



| Required Declarations |



| Inspection Requirement |



| Compliance / Renewal Requirement |



DOCUMENTATION GUIDANCE ENGINE

13. FOR EACH SERVICE, SYSTEM IDENTIFIES DOCUMENT REQUIREMENTS

Document types:

- Master business documents
- Land documents
- Project documents
- Technical documents
- Previous approvals
- Professional certificates
- Declarations
- Conditional documents
- Inspection-stage documents

For every document system knows:

- Required / Conditional / Not Required
- Why it is required
- Which approval requires it
- Whether it already exists
- Whether it is reusable
- Issue date
- Expiry date
- Valid / Expired / Missing
- Dependency on another approval

14. DOCUMENT UNDERSTANDING / HELP

User can understand:

- What is this document?
- Why is it required?
- What should it contain?
- Which regulation / GR requires it?
- Where can it be obtained?

Help is powered through the regulatory RAG assistant.

No separate complex rule system is required for plain-language guidance.

15. CONDITIONAL DOCUMENT LOGIC

Example:

Hazardous Waste = YES

↓

+ Hazardous waste details

+ Storage information

+ Relevant technical document

Hazardous Waste = NO

↓

These requirements disappear

Boiler = YES

↓

Boiler-specific documents appear



16. AUTOMATIC DOCUMENT REUSE

Example:

CTE approved

↓

CTE certificate automatically stored

↓

Issue date + expiry + source stored

↓

When CTO / other service needs CTE

↓

Existing certificate reused automatically

User does NOT upload the same approval repeatedly.



17. FORM DEPENDENCY CHECK

Example:

Form 14A

└─ Requires Form 14B

└─ Requires Annexure C

Main form cannot be submitted until required dependent forms are complete.





18. APPROVAL DEPENDENCY CHECK

Example:

Land Approval



Building Approval



Construction



Final Fire



CTO / Operating Approvals

Independent approvals are identified for PARALLEL processing.



19. BUSINESS JOURNEY / DEPENDENCY GRAPH

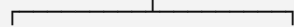
Tracks:

- What can start now
- What is blocked
- What is in progress
- What requires entrepreneur action
- What is approved
- What is rejected
- What is conditional
- What dependency is blocking downstream services
- Which services can run in parallel
- Which stage the business has reached



ESTABLISHMENT / PRE-CONSTRUCTION

LAND ROUTE



MIDC ROUTE



NON-MIDC ROUTE



MIDC land workflow / existing plot Land-use / planning / jurisdiction-specific route

LAND STAGE COMPLETE

20. MPCB - CONSENT TO ESTABLISH (CTE)

- Applicable service unlocked
- Common project data reused
- MPCB-specific information requested
- Documents attached
- Risk-based scrutiny later performed on department side

|| GENERIC APPROVAL PROCESS ENGINE ||

CTE APPROVED / RESOLVED

21. BUILDING / PLANNING STAGE

- Building Plan Approval
- Provisional Fire NOC
- Combined where applicable
- MIDC / relevant local planning authority

POWER

WATER

CONDITIONAL NOCs

- Boiler
- Sector NOC
- Hazardous activity
- Product-related
- Metrology

└ Fire-specific
└ Other conditional

PARALLEL WHERE POSSIBLE



22. UTILITIES – MINIMAL BUT INCLUDED

- Power
- Water

Utilities remain visible in the business journey without requiring your prototype/system to recreate full external utility backends.

Status can still be tracked as:

- Not Started
- Ready
- Submitted
- Pending
- Connected



23. CONSTRUCTION / INSTALLATION

- Building construction
- Plant installation
- Machinery installation
- Fire system installation
- Pollution-control installation
- Safety systems



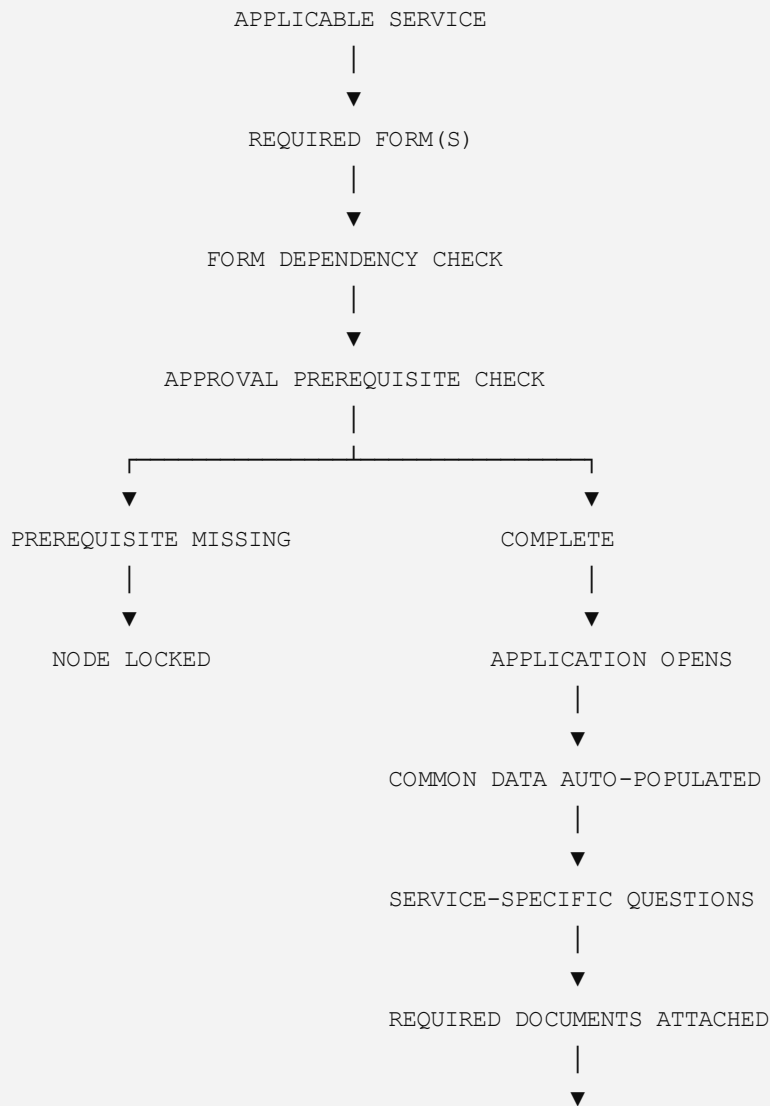
GENERIC APPROVAL ENGINE

APPROVAL / NOC / REGISTRATION NODE



APPLICABLE DEPARTMENT





FRONT-END PRE-VALIDATION

System objectively checks:

- All mandatory fields?
- All mandatory documents?
- Correct data format?
- Valid dates?
- Certificate expiry passed?
- Dependent forms complete?
- Required declaration accepted?
- Numeric values sensible?
- Basic calculations correct?
- Contradictory answers?
- Common data altered unexpectedly?
- Conditional requirement fulfilled?
- Required previous approval present?

System compares digital data across applications:

Master Profile:

Production = 100 T/day

MPCB Form:

Production = 100 T/day ✓

DISH Form:

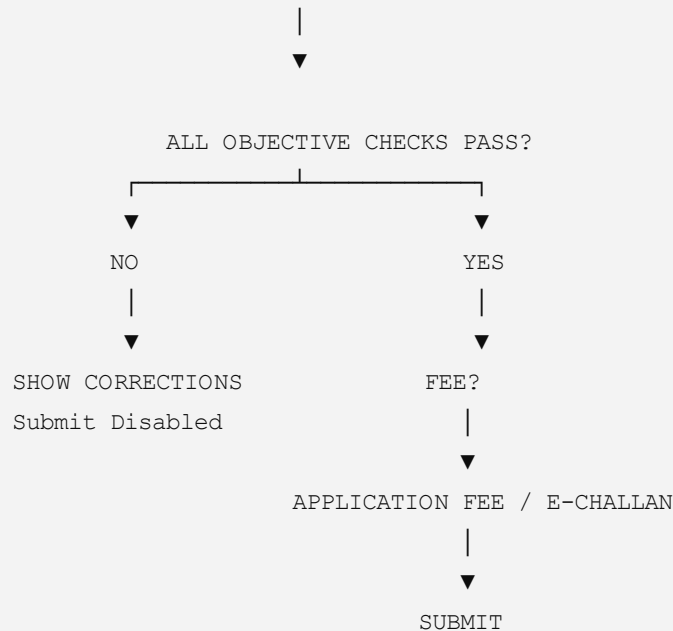
Production = 100 T/day ✓

Fire Form:

Production = 80 T/day ⚠ MISMATCH

Other cross-checks:

- Plot area
- Investment
- Employees
- Building area
- Water requirement
- Power
- Capacity
- Project location
- Company identity



===== GOVERNMENT SIDE =====

Application automatically reaches:

- Relevant department
- Relevant regional / district office
- Relevant service queue
- Relevant officer / desk where configured

Same common government system,
different department-specific parameters.



DESK-BY-DESK TRACKING

Application lifecycle may include:

Submitted



Fee / Challan



Document Desk



Initial Scrutiny



Technical Scrutiny



Inspection



Senior / Final Decision

System tracks:

- Current desk
- Time spent at current desk
- Previous desks
- Department time
- Applicant-hold time
- Inspection waiting time
- Total SLA elapsed



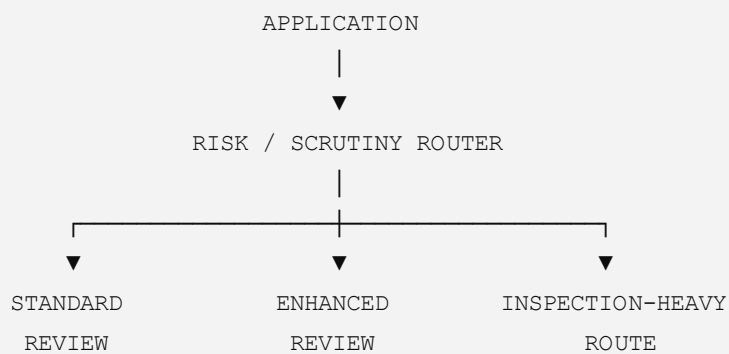
AUTOMATED PRE-CHECK

Department receives system summary:

- Form completeness
- Cross-form mismatches
- Expired certificate warnings
- Missing data
- Basic consistency
- Risk indicators
- Previous approval dependencies



Risk routing occurs mainly on department side.



Examples:

MPCB factors:

- Pollution category
- Effluent
- Air emissions
- Hazardous waste
- Chemicals
- Capacity

DISH factors:

- Hazardous process
- Workers
- Dangerous machinery
- Pressure vessels
- Chemical storage

Fire factors:

- Building type
- Building area / height
- Occupancy
- Hazardous storage
- Fire load

IMPORTANT:

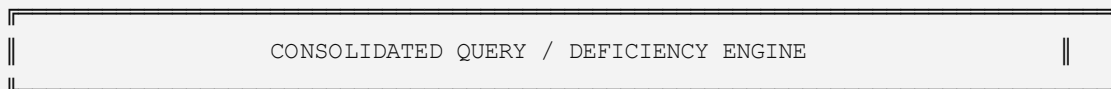
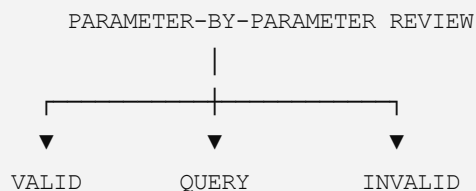
Risk determines HOW deeply the application is checked.

Risk does NOT automatically approve or reject it.



Officer checks:

- Data validity
- Document validity
- Certificate validity
- Legal acceptability
- Technical adequacy
- Consistency
- Authenticity where required
- Applicability
- Approval conditions
- Need for inspection



Instead of:

Query 1



Reply



Query 2



Reply

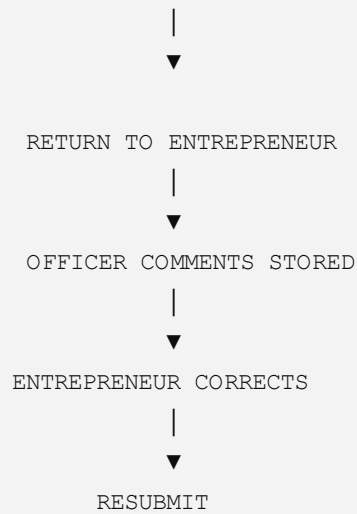


Query 3

System identifies unresolved issues together:

- Water balance mismatch
- ETP capacity mismatch
- Waste plan missing

Officer sends ONE consolidated deficiency memo.



DELTA RE-SCRUTINY

On resubmission, officer does NOT re-read everything.

System shows:

RESUBMISSION #2

Changed:

- Water: 50 → 65 KL
- ETP Capacity: 55 → 70 KL
- ETP Document v1 → v2

Unchanged:

- 42 other fields

Dependency impact:

Water changed

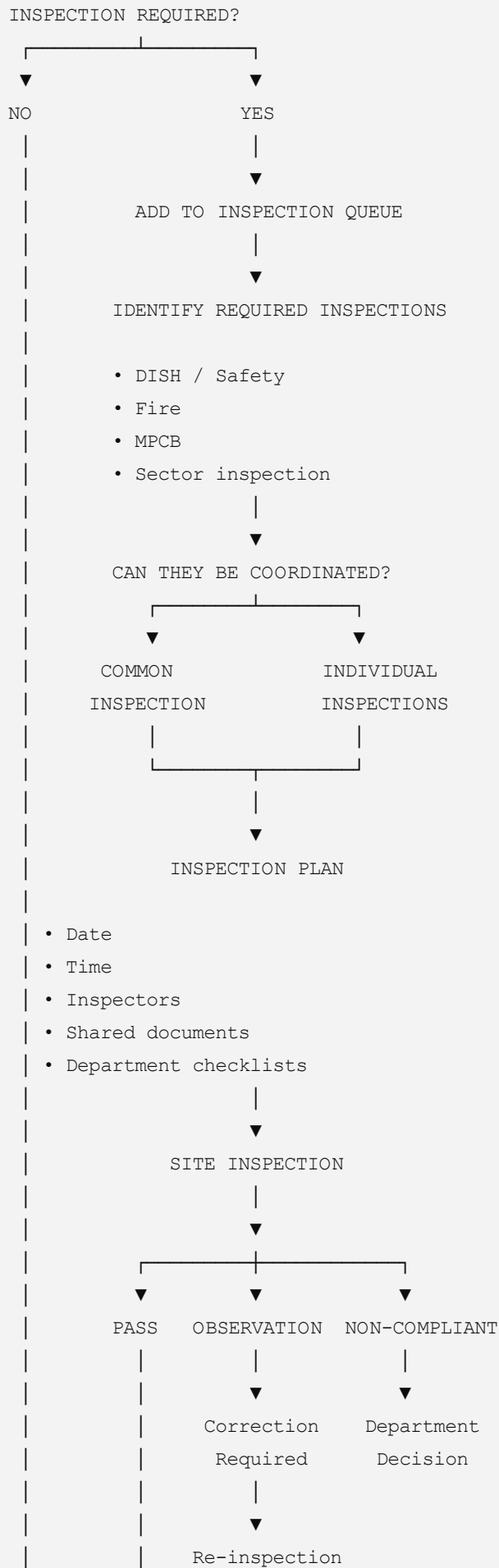
↓

Also re-check:

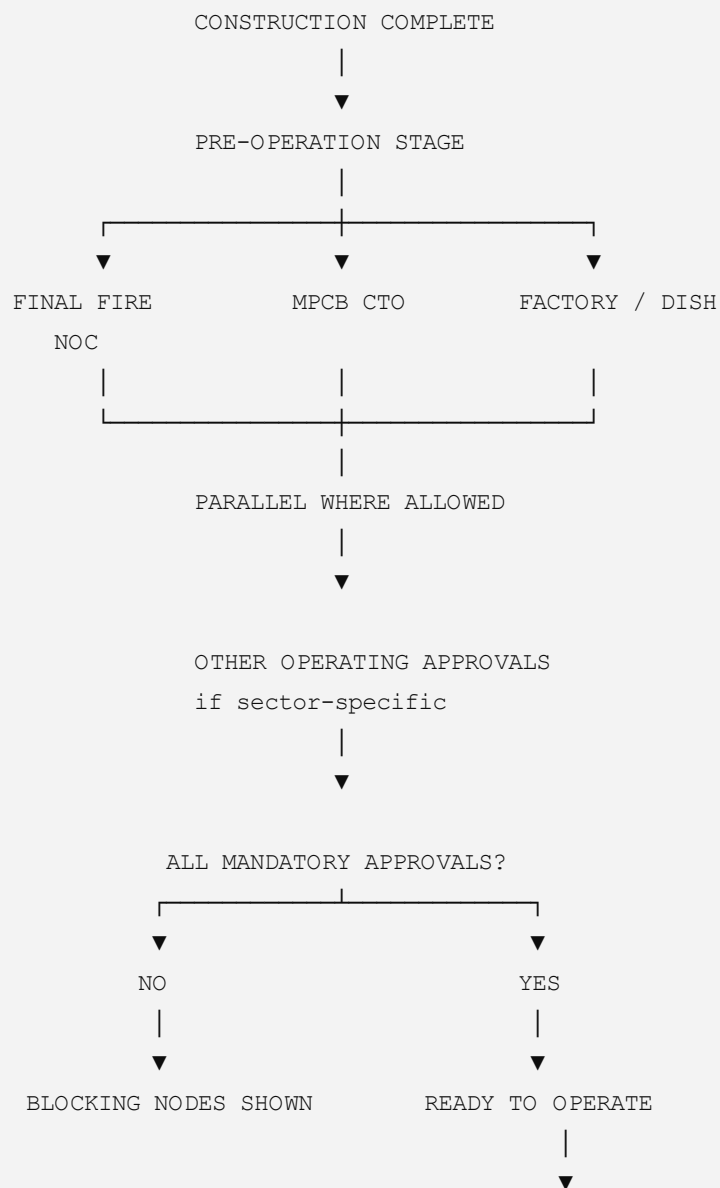
- Effluent
- ETP
- Related pollution parameters

Only changed + affected fields are prioritised for re-review.





CONSTRUCTION / PRE-OPERATION



OPERATIONS

BUSINESS STARTS OPERATIONS



APPROVAL → COMPLIANCE AUTO-GENERATION

When approval is granted:

APPROVAL ORDER



System records:

- Approval validity
- Renewal requirement
- Reporting requirement
- Periodic return
- Inspection requirement
- Approval conditions
- Special conditions



Compliance obligations are created automatically.

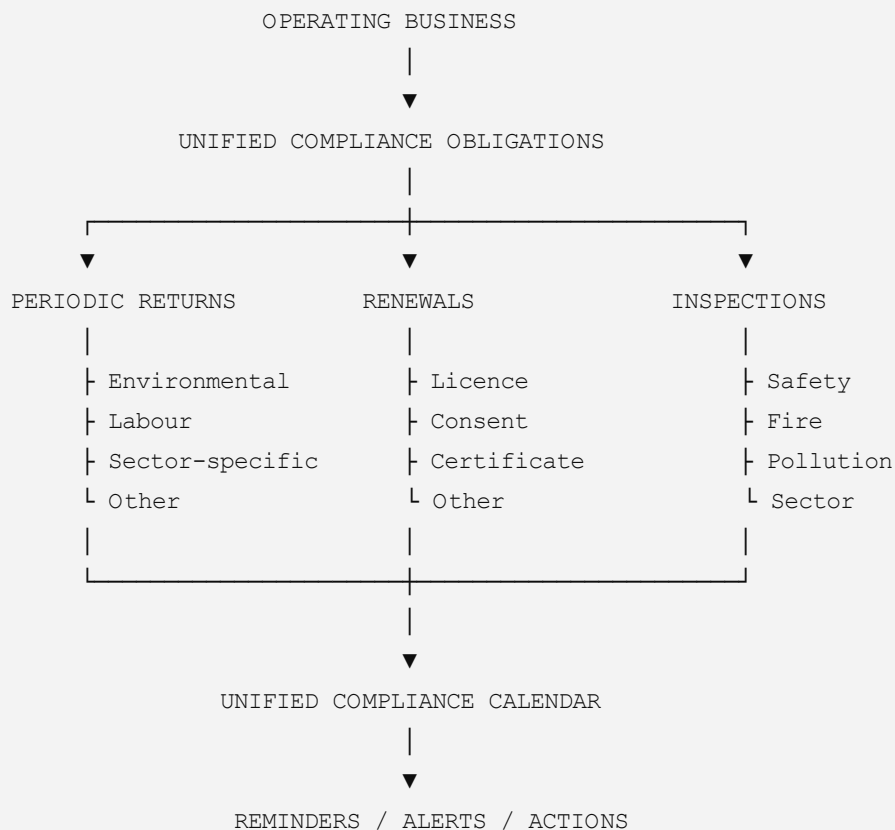
Example:

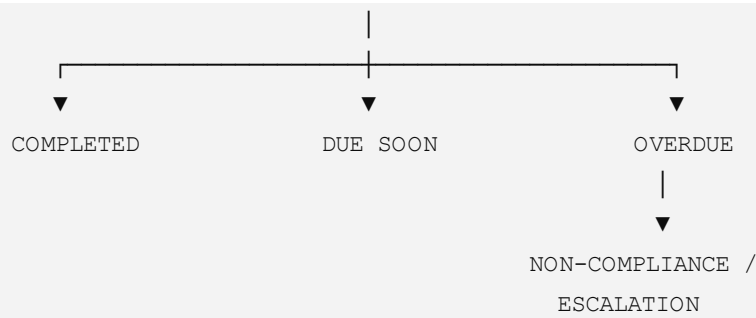
MPCB CTO APPROVED



Generate:

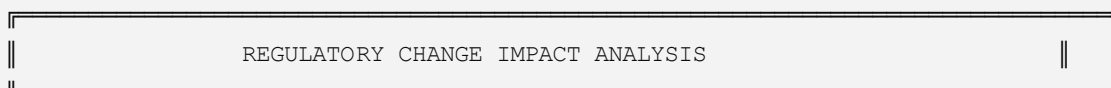
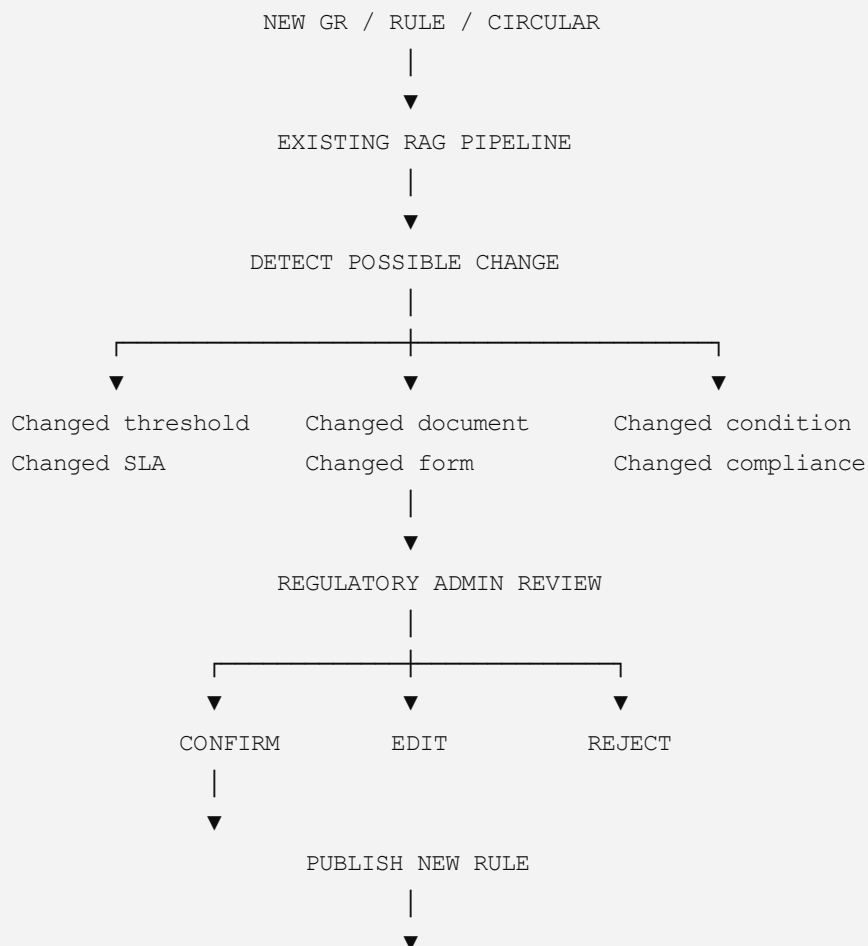
- CTO expiry
- Renewal window
- Environmental return
- Monitoring obligation
- Inspection condition
- Specific operating requirements





Compliance statuses:

- Compliant
- Due Soon
- Overdue
- Action Required
- Under Verification



System determines impact on:

- Active businesses
- Applications under preparation
- Submitted applications
- Existing approvals
- Upcoming renewals
- Compliance obligations
- Required documents
- Department procedures

Example:

NEW MPCB CIRCULAR

Affected:

- 37 operating businesses
- 11 draft applications
- 4 submitted applications
- 22 future renewals

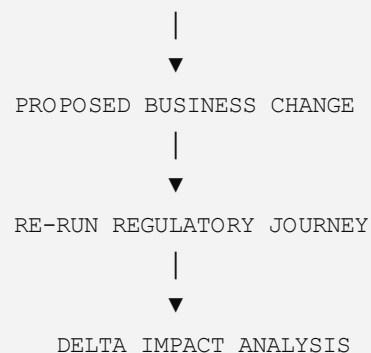
System:

- Updates requirement
- Alerts affected businesses
- Alerts affected officers
- Records rule version history



Operating entrepreneur proposes:

- Increase production
- Add machinery
- Add boiler
- Add chemical process
- Expand building
- Acquire additional land
- Change product
- Increase workers
- Change manufacturing activity



System identifies:

- Which approvals remain valid
- Which approvals need amendment
- Which new approvals are triggered
- Which inspections are triggered
- Which compliance obligations change
- Which incentive conditions may change
- Which documents need updating

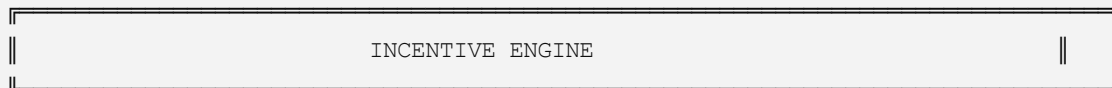
Example:

Production:

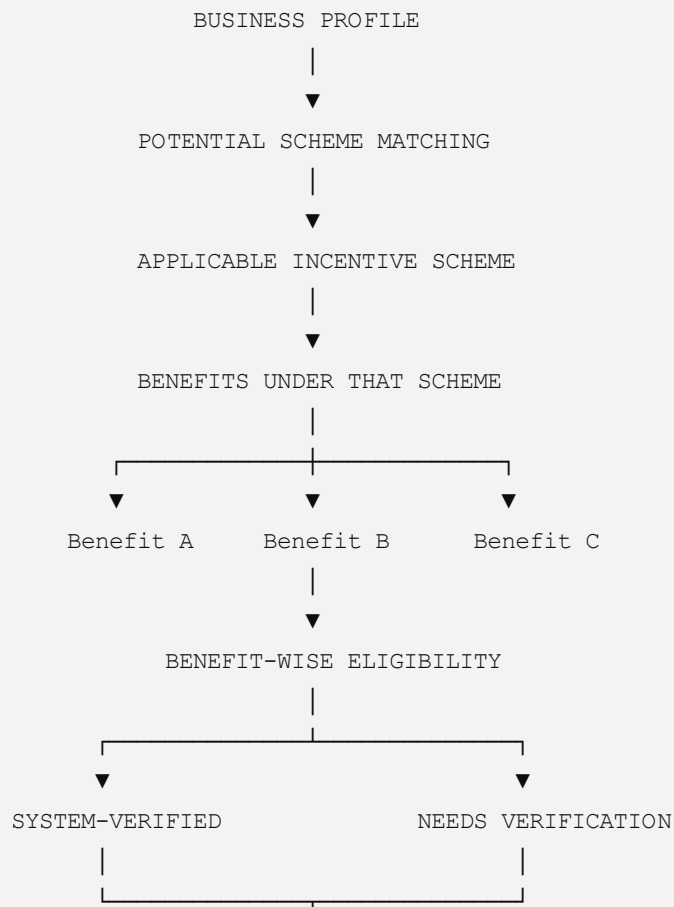
100 T/day → 200 T/day

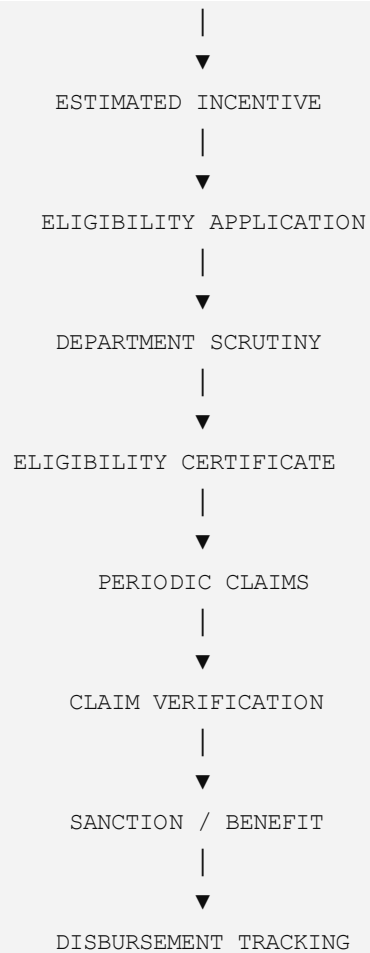
Impact:

- MPCB CTO amendment
- Factory licence update
- Possible inspection
- Fire unaffected
- Land unaffected



Runs in parallel from Business Profile stage.



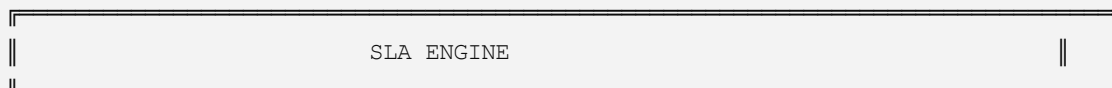


Important:

Scheme ≠ individual benefit.

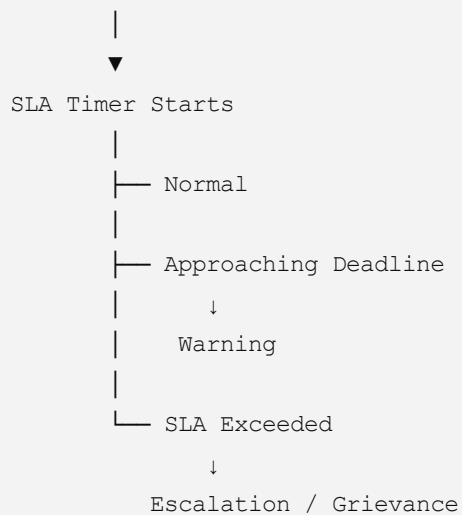
One scheme may contain multiple incentive benefits.

Eligibility can be assessed benefit-by-benefit.



Runs across every application and desk.

Application Submitted



Tracks separately:

- Department processing time
- Entrepreneur response time
- Current-desk waiting time
- Inspection waiting time
- External dependency waiting time
- Total elapsed time

Example:

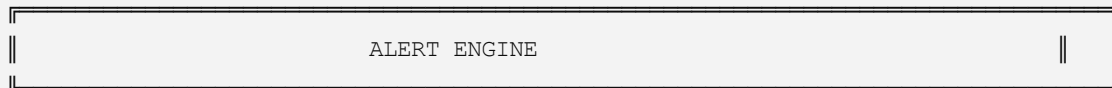
MPCB CTE

Technical Scrutiny

Day 18 / 21

Department processing: 13 days

Applicant response: 5 days



Alerts include:

- Missing information
- Missing document
- Expiring certificate
- Query raised
- Correction required
- Resubmission accepted
- Approval granted
- Approval rejected
- Inspection scheduled
- Inspection observation
- SLA at risk
- SLA breached
- Dependency unlocked
- Renewal due
- Compliance due
- Compliance overdue
- Scheme eligibility action
- Claim due
- Regulatory change affecting business
- Business change impact result



PROBLEM OCCURS

APPLICATION-LINKED GRIEVANCE

System automatically attaches:

- Application ID
- Department
- Service
- Current desk
- Submission date
- SLA
- Query history
- Applicant response time
- Department processing time
- Inspection status
- Previous escalation history

Grievance reasons:

- SLA breach
- Unresolved query
- Department delay
- Incorrect status
- Inspection delay
- Other

RELEVANT NODAL / OFFICER LEVEL

ESCALATION LEVEL

RESOLUTION

JOURNEY STATUS UPDATED

OFFICER REGULATORY RAG

During scrutiny officer can ask:

- Why is this parameter being checked?
- Which GR applies?
- What clause defines this requirement?

- What evidence is expected?
- Has this rule changed?
- What is the latest circular?
- Explain in Marathi / English.

RAG returns:

- Relevant source
- Relevant clause
- Effective date
- Related circular
- Supporting documents

IMPORTANT:

RAG assists the officer.

It does NOT approve / reject the application.

GOVERNMENT ANALYTICS / PROCESS MINING

System continuously analyses:

- Applications pending by department
- Applications pending by desk
- Average processing time
- Median processing time
- SLA breaches
- Query frequency
- Number of query rounds
- Common missing documents
- Common inconsistencies
- Common rejection reasons
- Common correction reasons
- Inspection waiting time
- Officer workload
- Department workload
- Applications stuck in rework loops
- Approval dependencies causing delays
- Geographic bottlenecks
- Sector bottlenecks
- Compliance violations
- Renewal delays
- Common inspection opportunities
- Grievance patterns
- Incentive claim delays

CAUSAL BOTTLENECK ANALYTICS

Not only:
"500 applications pending"

But:

MPCB CTE
Average journey: 24 days

Breakdown:

Applicant preparation	4.2 days
Document scrutiny	2.8 days
Technical scrutiny	6.1 days
Applicant response	3.5 days
Inspection waiting	5.9 days
Final decision	1.5 days

PRIMARY BOTTLENECK:
Inspection scheduling

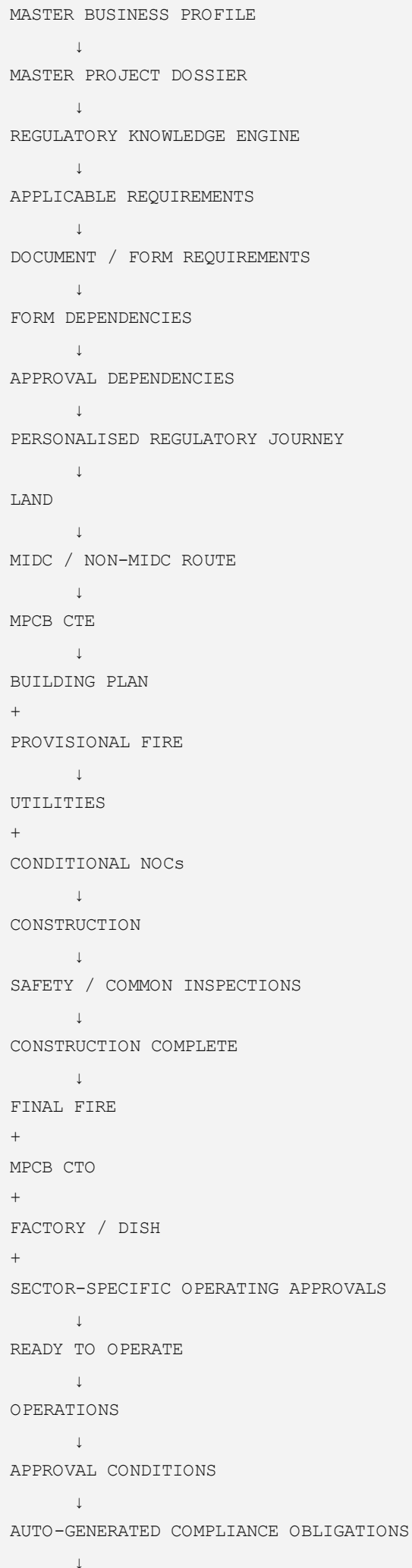
Responsible for:
27% of total processing delay

Example department insights:

1. Land Approval
Average delay: 11 days
Blocks 38 downstream approvals
2. Inspection Scheduling
24 applications waiting
3. Site Plan Issues
Cause 21% of correction requests
4. Officer Workload
Ratnagiri queue overloaded
5. Repeated Queries
Water-balance issue appears in 28% of MPCB cases



REGISTRATION
↓
CREATE PROJECT
↓
BUSINESS DISCOVERY
↓



RETURNS

+

RENEWALS

+

INSPECTIONS

↓

CONTINUOUS COMPLIANCE MONITORING

↓

BUSINESS EXPANSION / MODIFICATION

↓

REGULATORY IMPACT ANALYSIS

↓

AMENDMENTS / NEW APPROVALS

↓

CONTINUOUS BUSINESS LIFECYCLE

PARALLEL TRACK 1 – INCENTIVES

BUSINESS PROFILE

↓

SCHEME DISCOVERY

↓

BENEFIT-WISE ELIGIBILITY

↓

ESTIMATED INCENTIVE

↓

ELIGIBILITY APPLICATION

↓

ELIGIBILITY CERTIFICATE

↓

PERIODIC CLAIMS

↓

BENEFIT DISBURSEMENT

↓

TRACKING

PARALLEL TRACK 2 – REGULATORY INTELLIGENCE

NEW GR / RULE / CIRCULAR

↓

RAG DETECTION

↓

ADMIN VALIDATION

↓

RULE UPDATE

↓

IMPACT ANALYSIS

↓

AFFECTED BUSINESSES /
APPLICATIONS /

COMPLIANCES /
RENEWALS IDENTIFIED
↓
ALERTS + UPDATED JOURNEY

RUNNING THROUGHOUT THE ENTIRE SYSTEM:

MASTER DATA REUSE
+
VERIFIED DATA PROVENANCE
+
DOCUMENT REUSE
+
DEPENDENCY MANAGEMENT
+
CROSS-FORM CONSISTENCY
+
DESK-BY-DESK TRACKING
+
RISK-BASED SCRUTINY
+
CONSOLIDATED QUERIES
+
DELTA RE-SCRUTINY
+
COMMON INSPECTION PLANNING
+
SLA TRACKING
+

The most important conceptual upgrade compared with yesterday is that the platform is no longer just an **approval portal**. It now has four connected lifecycles:

1. BUSINESS SETUP LIFECYCLE
Discovery → Approvals → Inspection → Operation
2. REGULATORY LIFECYCLE
GR/Rule → Knowledge Engine → Impact → Updated Requirements
3. COMPLIANCE LIFECYCLE
Approval → Obligations → Returns/Renewals → Monitoring
4. GROWTH LIFECYCLE
Incentives + Expansion → Impact Analysis → Amendments

And importantly, I have **not added back the document-readiness score** you decided to skip. Documentation guidance remains based on required/conditional/reusable/expired/missing documents, dependency and RAG-based assistance—not an arbitrary percentage.

